

Voting-based seismic data classification system using logistic regression models

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Abstract—Early detection of seismic events is fundamental to mitigate the damages produced by these natural phenomena, and mainly, save lives. In this paper, logistic regression models were applied to distinguish between seismic and non-seismic local events considering data segments from four seismological stations (epicenter in the Santander department, Colombia) of the National Seismological Network of Colombia. The classification outputs obtained by each individual classifier were then combined using voting functions. For each station, Degree of Polarization, Ratio of Vertical power to Total power, Skewness and Kurtosis of the three-component seismic data were extracted. Results showed that the best individual classifier achieved an accuracy of 95% and reaches an accuracy of 98% using voting functions. This classifier is the first step in the development of alert mechanisms for the early detection of seismic events in Colombia.

Keywords— *Earthquake, seismic event classification, machine learning, logistic regression, voting-based system.*

I. INTRODUCTION

Natural earthquakes are sudden movements produced by vibrations caused by the relaxation and reactivation of the energy accumulated by the deformation of the lithosphere, which propagates through the stratum (geological layers) [1]. When occurring, earthquakes release seismic waves that affect the infrastructure and population on Earth. They cause also landslides, floods, tidal movements, and wildfires. There are several vulnerable areas around the world prone to the occurrence of disastrous seismic events, such as the Pacific Ring of Fire, that makes Chile, Mexico, Japan, Indonesia, and Colombia very susceptible to earthquakes [2].

Seismic events can be documented and stored as seismic records, consigning information about the signals registered. Depending on the region of interest, it is possible to obtain seismological records through web services, such as the Chilean Seismological Network and the Integrated Plate Boundary Observatory in Chile, the National Seismological Service of Mexico for Mexico and some Latin-American records, the Japan Meteorological Agency for information related to weather, climate, and seismic activity, the Meteorology, Climatology, and Geophysical Agency for information about climate and seismic activity at Indonesia, the

United States Geological Survey for the United States of America and other international records, the European-Mediterranean Seismological Centre for the countries around the Mediterranean, and the National Seismological Network of Colombia (RSNC) for Colombian seismic activity and around countries records. We analyzed the seismic history of events in the department of Santander, Colombia, considering four stations of the RSNC, a network of 85 stations of seismometers and accelerometers that records the national seismic activity.

The Colombian soil has complex geomorphology that is directly influenced by nearby plates and their continuous interactions [3] [4]. According to the Colombian Association of Seismic Engineering, 87% of the Colombian population is at seismic risk (40% at high threat level and 47% at moderate threat level) [5]. In particular, the Mesa de Los Santos region (Santander department), which is traversed by the Santa Marta – Bucaramanga fault, is considered as the second most active earthquake zone in the world [6]. Highly destructive seismic events in Colombia, such as the Quindío earthquake in 1999, which caused 1200 deaths, show the overall concern about the damage implications related to the Colombian seismic activity [7] [8]. Consequently, it is necessary to plan, prevent and mitigate the effects of natural disasters of the seismic category using early detection mechanisms that allow the identification of the seismic waves, among which are the P-waves (for primary). These waves mark the beginning of upcoming earthquakes, having special characteristics regarding frequencies and magnitudes.

The goal of this paper is to present an offline binary classification system based on logistic regression to distinguish between seismic and non-seismic (noise, non-natural earthquakes, or artificial-source earthquakes) events. The classification was done by considering segments of time series taken from the RSNC dataset. Four logistic regression classifiers were applied to the data from four stations, one classifier for each station, in order to produce independent decisions about the seismic class. Then, a voting function was used to combine the decision of each classifier (i.e. a decentralized fusion scheme). We used logistic regression classifiers since the logistic regression model is the simplest

technique to use for this type of problem and has already been used on similar seismic records [14] [15].

II. RELATED WORK

A state-of-the-art study was done to gain insight into the machine learning techniques for the detection of seismic events, the common features considered, the number of stations used and their geographical distribution. Machine learning techniques have also been used in similar tasks, such as for tomographic analysis of soil [9], detection of volcanic events [10] and the specification of technological infrastructure for the analysis of seismic records [11].

In Table I we have detailed the state-of-the-art relevant to our work. We have highlighted the technique used and the number of stations considered for the classification process, as well as the training and testing datasets, the pre-processing and features extraction processes and the accuracy obtained. From this table we can observe that an accuracy between 98% and 99.7% has been obtained in China and Southeast California using logistic regression, an accuracy between 81% and 85% has been obtained in the United States using Support vector machines (SVM), Random Forest, and Bayesian approaches, and an accuracy between 90% and 99% have been obtained in India, Morocco, and Italy using Artificial Neural Networks and Multilayer perceptron networks.

III. METHODOLOGY

The methodology followed in this work is shown in the block diagram of Fig. 1. Each of the stages is detailed in the sections below.

A. Seismic dataset selection

The seismic records analyzed were obtained from the RSNC service (data available at <https://www2.sgc.gov.co/sismos/>). These are composed of seismic information files and content files with samples of the events.

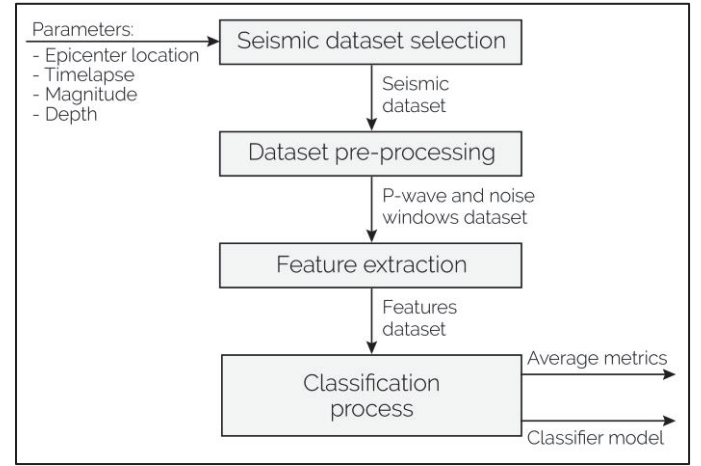


Fig. 1. Block diagram of the methodology carried out in this work.

The dataset consisted initially of 60,785 local seismic events with an epicenter in the department of Santander, Colombia, between January 1, 2010, and September 30, 2017. Each of these events is composed by the signals recorded by 85 national's seismological RSNC stations. Each station measured the ground speed in several spatial components, with a topological dimension between 1 and 16, with 1 and 2 rare cases and more than 9 isolated cases. The topological dimension considered is a 3-component dimension: Z, N and E components. Each component recorded the seismic event between a seismic window that includes samples before and after the event.

TABLE I. STATE OF THE ART OF SESMIC DATA CLASIFICATION RELEVANT TO OUR WORK.

Method	Ref	Place	Triaxial stations	Training and Test Sets	Data pre-processing and Features extracted	Accuracy (%)
Logistic Regression	[14]	China	1	Training (80%): 327 events Test (20%): 82 events Total events: 409 From 1997 to 2017	Features: not reported.	98 to 99.7
	[15]	United States	13	Training (60%): 12,453 events Test (40%): 8,302 events Total events: 20,755 From 2015 to 2016	Bandpass filter 0.5-49.9 Hz. Features: DOP and frequency-averaged spectrograms.	99
SVM, Random Forest and Bayesian Frameworks	[16]	United States	1	Training (75%): 426 events Test (25%): 142 events Total events: 568 From 1970 to 2017	Features: Gutenberg-Richter laws a-value, mean square deviation, magnitude deficit, elapsed time, mean time, coefficient of variation, rate of square root of seismic energy and mean magnitude.	81
	[17]	United States	16	Training: 86,400 samples/day for 3 months.	Features: station location and channels, event's start time, end time, number of sample points, maximum amplitude of the wave and standard Deviation.	85
Artificial Neural Networks and Multilayer perceptron (MLP) network	[13]	India	1	Training (38%): 60 events Test (62%): 100 events Total events: 160	Bandpass filter 1-8 Hz and 2-s sliding windows. Features: Degree of Polarization (DOP), Auto Regression Coefficient (ARC), STA/LTA ratio and Ratio of Vertical power to Total power (RV2T).	90 to 95
	[18]	Morocco	5	Testing: 343 events	Features: envelope similarity, event's time duration, spectral Centroid, spectral length, and skewness.	94
	[19]	Italy	1	Training (63%): 315 events Test (37%): 189 events Total events: 504	Features: not reported.	97.7 to 98.5

Of the 60,785 seismic records, numerous files were discarded because of illegibility (4,629 files), repeatability (2,306 files), wrong format (1,850 files), incompleteness (3,698 files) and sensor variation (33,085 files). At the end of the selection process, 10,288 from the 14,947 events of the dataset were selected, in which four seismological stations were chosen: BRR, RUS, PTB, and PAM, considering their epicentral distance relative to the events, sorted from lowest to highest. To avoid class imbalance, the relation between seismic and non-seismic labeled events was 50%.

The reading and interpretation of the records were done using the Python language and the algorithms developed for this purpose, through the Obspy, Numpy, SciPy, Pandas and Scikit-Learn libraries.

B. Data pre-processing and feature extraction

The first stage of the preprocessing was the filtering of the data. For this process, two main sources of noise were considered, the terrestrial vibrations that have natural causes, and those in which the instrument of measurement is involved. A fourth-order Butterworth bandpass filter of 1-10 Hz was applied since the seismic signals do not contain useful information outside this range of frequency: below 1 Hz are micro-seismic components and ground noises, and above 10 Hz are natural and man-made noise sources [12].

Once filtered, the signals were normalized by feature scaling, a process that centralizes the samples and allows to obtain a dataset without any continuous component, with a re-scaled magnitude by maximum and minimum. Because the samples have a high diversity of sampling rates, product of the different sensors and acquisition devices, the digital conversion, the storage, and the processing algorithms, signals were resampled to 30 Hz. This sample frequency was selected according to the signal content after the filtering process applied and is, indeed, greater to the Nyquist frequency.

Fig. 2 depicts an example of a seismic signal before (in blue) and after (in green) the pre-processing stage. The original amplitude is on another scale due to the output of the analog-to-digital process done by the RSNC. Subsequently, a 60-sample window was extracted from the three components recorded for each event and the named stations, in such a way that P-waves were at 50%.

Conversely, the classification of seismic events was carried out through the recognition of various signal features. For the three-component seismic input dataset, four attributes were identified for this analysis: *i)* Degree of Polarization (DOP), calculated three-dimensional, *ii)* Vertical Power Radius against Total Power (RV2T), calculated in the three components, *iii)* skewness, calculated in the three components, and *iv)* Kurtosis, calculated in the three components. DOP and RV2T have proven to be useful for the classification of seismic records [12].

C. Seismic data classification

An offline binary classification of time series segments was carried out, considering two output classes, positive and negative. On one hand, observations labeled as positive class exhibited seismic features and hence were cataloged as

earthquakes; on the other hand, observations labeled as negative class had noisy or non-natural seismic features in their nature, therefore were labeled as noise (non-natural local earthquakes, or artificial-source local earthquakes).

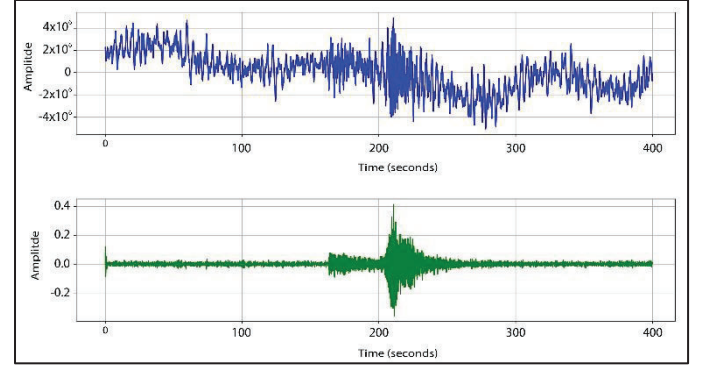


Fig. 2. Example of a signal before (in blue) and after (in green) the pre-processing stage.

The classification process is shown in Fig. 3 and was done using four logistic regression models, each one related with each of the named seismological stations and considering the equation (1), where x_i represents the features extracted and β_i is the coefficient that accompanies each x_i . Once the data was obtained from those four channels (one per station), it became an entry (X) for the logistic regression models, which are trained for classifying the events supplied.

$$\text{logit}(p) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_8 x_8 \quad (1)$$

The classification decision of each model (Y) is then combined into three decentralized functions: AND, OR and Voting functions. These functions compare the detector's decisions with each other. For the AND function, four decisions must be positive to classify the event as positive. In the same sense, the OR function needs only one decision to be positive to classify the event as positive, and the voting function needs at least half inclusive of the decisions to be positive, to throw a positive result.

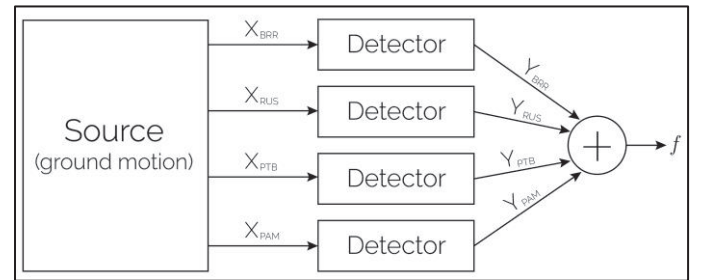


Fig. 3. Seismic data classification based on decentralized data function.

D. Performance evaluation

Cross-Validation of 10 iterations was performed to achieve some confidence in the classification results for the logistic regression of individual models and assure the reproducibility of the results. In this sense, for each iteration, the datasets were randomly assigned to two subsets: 80% for training and 20% for testing. Then, the logistic regression models were combined in the decentralized functions and cross-validation of 10 iterations

was again performed to achieve some confidence in the inclusion of decentralization functions.

The performance of the regression models and decentralized functions was measured using accuracy. This metric was computed considering the relation between true positives, true negatives, false positives, and false negatives, extracted from the outgoing confusion matrices.

IV. RESULTS

The outgoing performance metrics of the logistic regression models and the decentralized functions implemented are shown in Fig. 4. From this figure, we can see that the PTB station provides the highest accuracy whereas the BRR station provides the lowest accuracy. Moreover, thanks to the combination of the four detectors, the accuracy reaches 98.26% using the voting function.

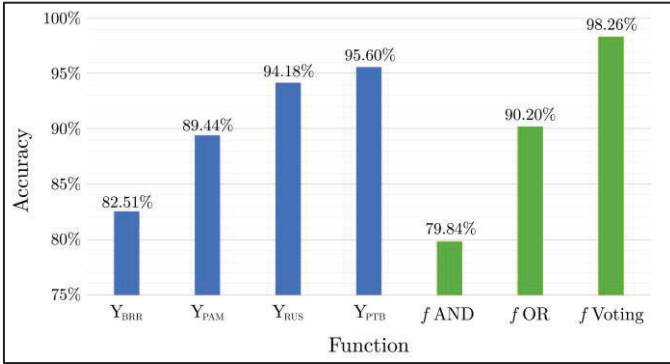


Fig. 4. Classification performance for the individual decision and the decentralized fusions.

The regression coefficients of the models with the highest accuracy on each of the 10 cross-validation iterations per seismological station are shown in Table 2.

TABLE 2. REGRESSION COEFFICIENTS FOR THE HIGHEST ACCURACY MODEL OF THE CROSS-VALIDATION PROCESS PER SEISMOLOGICAL STATION.

Regression coefficient	x_i	Function			
		Y_{BRR}	Y_{PAM}	Y_{RUS}	Y_{PTB}
β_0	-	1.0264	1.5928	1.9634	2.3979
β_1	DOP	-0.5176	0.0781	-0.8519	0.1192
β_2	RV2T	0.9323	1.1792	1.8836	0.7753
β_3	Kurtosis Z	2.9005	3.4008	3.6233	3.9034
β_4	Kurtosis N	0.1730	0.2089	0.6635	1.4677
β_5	Kurtosis E	0.2389	0.7747	0.5933	0.9165
β_6	Skewness Z	-0.0889	0.0209	-0.0841	0.1102
β_7	Skewness N	0.0044	-0.0041	-0.0545	-0.0272
β_8	Skewness E	0.0236	0.0793	-0.1041	0.0050

V. DISCUSSION

The significance of this study lies in the application of logistic regression models to classify seismic data, acquired from four distinct seismological stations and using decentralized functions framed in the voting system developed. Also, the logistic regression models show a performance that surpasses the 80% accuracy individually and one of them is slightly above 95%. It means that considering just one triaxial station, it can be achieved a performance that outstrips the initial expectations. As

the regression coefficients showed, the four models were totally independent and different from each other. Furthermore, when combining the four models in a voting function, the performance increases by almost 3%, showing an accuracy greater than 98%.

The differences found between the individual performances of the classifiers could have been caused by some unexpected variations in the epicentral distances of the stations observed in the seismic events and some other seismically factors like the topography and the stratum heterogeneity. The diversity of associated electronic devices in the seismological stations and the variety of the acquisition, transmission, and data processing algorithms could have also been a source of bias that affects the data classification performance.

Nevertheless, there were a few practical difficulties that should be highlighted. One challenge was to use all samples identified in the sample selection stage, though there were storage and processing mistakes from the source that made this impossible. The use of logistic regression models to identify non-linear nature seismic signals is a less robust technique. Using more robust techniques like Artificial Neural Networks, SVM or Bayesian approaches could be carried out to identify in a better way the non-linear nature of the seismic signals; our research team is currently working on this direction.

VI. CONCLUSIONS

The results of this work showed that using logistic regression models and a voting system approach, it was possible to automatically classify seismic events in two classes based on seismic and statistical variables, namely, Degree of Polarization, Vertical Power Radius against Total Power, Kurtosis and Skewness. The classification accuracy is over 98% using a four-triaxial-stations seismic data from the National Seismological Network of Colombia with the epicenter in Santander.

In accordance with the results presented, the following improvements and future work are proposed: enlarge the database to include other national events besides the ones within Santander location, strengthen the seismological background to incorporate new seismological stations, apply more robust Machine Learning techniques for classifying seismic events, include other centralized and decentralized functions identify additional features that could provide a better description of the nonlinear dynamics of seismic signals. This would help in the early detection of seismic events and the developing of future alert mechanisms.

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